



TANISHA NATALIA ARITONANG

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Medan, North Sumatera

Computer Engineering student with strong foundations in IT infrastructure, Linux administration, networking, and DevOps tools. Experienced in deploying containerized applications and automating workflows, with additional capability in AI model integration and modern web technologies.

Education

Institut Teknologi Del - Sitoluama, kecamatan Laguboti, kabupaten Toba Samosir	Aug 2023 - Sep 2026 (Expected)
<i>Diploma in Computer Engineering, 3.80/4.00</i>	
Academic Excellence Scholarship – Institut Teknologi Del (2023 – 2024) Awarded a partial tuition scholarship for maintaining a high GPA (3.80) and outstanding academic performance.	

SMAN 1 Girsang Sipangan Bolon - Parapat	2020-2023
Senior High School, Science	

Work Experiences

Institut Teknologi Del	Aug 2024
<i>Matriculation Assistant</i>	
- Assisted in facilitating the matriculation program for approximately 500 new students by supporting academic sessions, managing class attendance, and helping students adapt to the university environment. - Provided basic instruction in subjects such as introductory programming, Microsoft Word, and other fundamental digital literacy skills, while also coordinating with lecturers and ensuring smooth classroom activities.	

Institut Teknologi Del	Sep 2025
<i>Vice Chair of KPU HIMATEK (Election Commission of the Computer Engineering Student Association)</i>	
- Assisted the Chair in leading the HIMATEK Election Commission (KPU HIMATEK), responsible for organizing and supervising the student association election process. - Coordinated the planning, timeline, and procedures for transparent and fair elections of HIMATEK leaders. - Managed communication between candidates, election committees, and student representatives. - Ensured election integrity by implementing fair voting regulations and documentation processes.	

Organization

Himpunan Mahasiswa Teknologi Komputer (HIMATEK) – Insitut Teknologi Del	Aug 2024 - Present
<i>Leader of Competition Division</i>	
A student-led organization at Institut Teknologi Del that focuses on academic development, competitions, and professional growth of Computer Engineering students through training, workshops, and collaborative programs. - Led the Education Division and initiated academic programs supporting technical skill development among students. - Coordinated internal workshops and collaborations with lecturers to enhance learning engagement. - Mentored junior members and promoted continuous learning culture within the association.	

Projects

AI-Supported Adaptive Learning System for Computer Networks (2025): Tech Stack: Python, JavaScript, Next.js, Docker, Hugging Face, PyTorch, PostgreSQL Developed a web-based adaptive learning platform using a fine-tuned LLaMA 3.1 8B model. Built modular SFT training scripts with LoRA, integrated wandb for monitoring, and containerized the system for cloud deployment. Chatbot for Education Platform using Analogy-Based Learning (2024): Tech Stack: OpenAI API, LangChain, Supabase, Docker Built an analogy-driven chatbot for educational exploration. Integrated LangChain for contextual reasoning and Supabase for semantic retrieval, deployed via Docker with a lightweight interactive web interface. Infrastructure Automation Project using Ansible (2024): Tech Stack: Ansible, Linux, Bind9, Postfix/Dovecot, HAProxy, Squid Proxy, Bash, VirtualBox. Automated deployment and management of a multi-server LAN environment using Ansible, including DNS & sub-domains, mail servers, load balancing, and forward proxy, enabling consistent provisioning and faster setup with reusable playbooks and roles. Website on the Use of Analogies in Parent-Child Communication (2024): Tech Stack: WordPress, Elementor Developed a WordPress-based educational site promoting analogy use in parent-child communication. Designed responsive layouts, published interactive content, and optimized usability and content management.	
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- **Embedded System - Automatic Laundry System (2024):**

Tech Stack: Arduino, C/C++, Servo Motor, Rain Sensor, LDR, Relay, DC Fan, LCD Display

Built an automatic laundry system with servo-controlled roof, dryer fan, LDR, and rain sensor for weather-adaptive operation.

Achievements & Skills

- **Achievements | PILMAPRES 2025 | Kementerian Pendidikan, Kebudayaan, Riset, dan Teknologi (2025)**
- **Achievements | Microsoft Certified: Azure AI Fundamentals** [🔗](#) (2025)
- **Achievements | AWS Cloud Practitioner Essentials** [🔗](#) (2025)
- **Achievements | Complete Linux Training Course | Udemy** [🔗](#) (2025)
- **Achievements | Go Programming Language | Great Learning** [🔗](#) (2025)
- **Achievements | HCIA-Storage V5.0 Course | Huawei** [🔗](#) (2025)
- **Achievements | HCIA-Datcom V1.0 Course | Huawei** [🔗](#) (2025)
- **Achievements | HCIA-Cloud Computing V5.5 Course | Huawei** [🔗](#) (2025)
- **Achievements | AI Agents Fundamentals | Hugging Face** [🔗](#) (2025)
- **Achievements | LLM Course Unit 3: Fine-tuning Language Models | Hugging Face** [🔗](#) (2025)
- **Achievements | Participation_Del Debate Competition 2025 British Parliamentary Debate with UPT Bahasa IT Del (2024)**
- **Soft Skills:** Analytical Thinking, Problem-Solving, Collaboration, Adaptability, Communication and Ownership
- **Hard Skills:**
 - IT Infrastructure & Sysadmin : Linux Administration (systemd, permissions, file system, networking), OS installation & configuration (Windows, macOS, Linux), Hardware & software troubleshooting, Networking (TCP/IP, VLAN, routing basics).
 - DevOps & Cloud : Docker, Docker Compose, Kubernetes (basic), Terraform (IaC), CI/CD (GitHub Actions), AWS (EC2, S3, IAM, VPC concepts)
 - Programming & Frameworks: Python, JavaScript, C, Next.js, Golang
 - Tools & Version Control: Git, GitHub, Shell Scripting, Weights & Biases (wandb)
 - AI & Machine Learning: Hugging Face, LangChain, OpenAI API, LLM Fine-Tuning (LoRA, SFT)
 - Data & Databases: PostgreSQL, Supabase, MySQL, Vector Databases, JSON/CSV Handling
 - Web Development: WordPress (Elementor, PHP), HTML, CSS, REST API Integration
- **Language:** English & Indonesia